Learnable Background Endmember With Subspace Representation for Hyperspectral Anomaly Detection

Tan Guo, *Member, IEEE*, Long He, Fulin Luo, *Senior Member, IEEE*, Xiuwen Gong, Lei Zhang, *Senior Member, IEEE*, and Xinbo Gao, *Fellow, IEEE*

***Abstract*— Hyperspectral anomaly detection (HAD) aims to label each hyperspectral image (HSI) pixel as background or anomaly, in a totally unsupervised manner. Thus, a fine background representation is vital to obtain good HAD performance. This article introduces background endmember representation and proposes a novel HAD method termed learnable background endmember with subspace representation (LEBSR). First, the HSI is unmixed to simultaneously obtain the background endmembers and their abundances. The three constraints of p-norm, sum-to-one, and nonnegativity work together to promote a more meaningful and accurate background endmember representation. In addition, a mapping matrix with orthogonality is jointly optimized to transform the priori backgrounds into the background endmember subspace, and then the mapped priori backgrounds are approximated to the low-rank representation (LRR) with the background endmembers. With the methodology, the backgrounds can be well reconstructed under the guidance of the priori background information to accurately detect anomalous pixels with the reconstructed residuals. The experimental results on several HSI datasets verify the superior performance of LEBSR than the state-of-the-art methods. https://github.com/HalongL/HAD-LEBSR**

***Index Terms*— Abundance estimation, hyperspectral anomaly detection (HAD), learnable endmember, subspace representation.**

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I. Introduction

HYPERSPECTRAL image (HSI) is a 3-D cube composed of hundreds of narrow-band images, which contains richer spectral information than multispectral images [1], [2], [3]. Different spectral information corresponds to different material properties, i.e., the spectral information can be used to reveal different material components [4], [5], [6]. Hyperspectral anomaly detection (HAD), as an important application of HSI, has been receiving continuous attention due to its important role in many tasks [7], [8], [9]. Anomalies in HSI are mainly distinguished from background pixels in two aspects: first, spectral differences, where there are obvious differences between the spectra of anomalous targets and their surrounding background pixels; second, spatial differences, where the background pixels are usually regions that are prevalent and uniformly distributed in the image, whereas anomalies are regions that are different from or stand out from their surroundings [10], [11]. Therefore, most of the existing methods start from spectral and spatial differences to construct HAD methods.

A classical HAD manner is based on statistical theory. These methods consider all pixels in HSI as high-dimensional vectors, and assume that the backgrounds obey a certain probability distribution, while the anomalies do not conform to the probability distribution. The Reed–Xiaoli (RX) detector is the most representative method, which uses a Gaussian model to estimate anomaly pixels based on the global mean and covariance of HSIs [12]. Instead, for local RX detection (LRX), a local window is designed to compute the background covariance matrix [13]. Following further development, some advanced statistical-based methods have been designed for AD of HSIs such as the support vector data description (SVDD) [14] method representing the local background of HSIs, kernel isolation forest detector (KIFD) [15] using the classical isolation forest (iForest) algorithm to accurately detect the anomaly in HSIs, and two-step generalized likelihood ratio test (2S-GLRT) exploiting the GLRT criterion with two-step [16].

Most of anomaly detection studies based on statistical theory assume that the backgrounds match a statistical

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distribution model. For the complexity of backgrounds in HSI, coupled with the interference of many factors, it is not easy to establish a statistical distribution model that matches the real hyperspectral scene, which limits the application of anomaly detection based on statistical theory. Representation-based HAD methods assume that the pixels belonging to the backgrounds can be well reconstructed by the background dictionary, while anomalous pixels cannot. And the reconstruction error is considered to be the degree of anomaly of the measured pixel. According to different regularization constraints imposed on the models, they can be further classified as low-rank representation (LRR) models [17], sparse representation (SR) models [18], and collaborative representation (CR) models [19], [20]. Among them, LRR is a globally constrained model that treats the original image matrix as the sum of the background matrix and the anomaly matrix. The backgrounds have global low-rank characteristics due to spatial homogeneity and spectral correlation, while anomalies are rare because of their small proportion and low probability of occurrence. Based on LRR theory, robust principal component analysis (RPCA) [21] decomposes the HSI into low-rank and sparse matrices by combining the low-rank and sparse properties in HSI. However, RPCA is sensitive to noise. To reduce the influence of noise on the detection results, some HAD methods consider the noise model in LRR including low-rank sparse matrix decomposition (LRaSMD) [22], and low-rank and sparse matrix decomposition-based mahalanobis distance for AD (LSMAD) [23], low-rank and SR (LRASR) [24], local spatial constraint and total variation (LSCTV) [25], and anti-noise hierarchical mutual-incoherence-induced discriminative learning (AHMID) [26], and matrix factorization with framelet and saliency priors (SaFra) [29]. Anomaly detection methods based on SRs adaptively select several atoms from an overcomplete dictionary to represent a pixel under test using the l_1-norm, such as the background joint SR detector [27] and archetypal analysis and structured SR [28]. Unlike SR, CR-based anomaly detection methods exploit the collaboration of atoms in the dictionary by using the l_2-norm to represent each pixel under test [19], such as CR-based detector (CRD) [30], and relaxed CR detector (RCRD) [31].

In recent years, deep learning has achieved promising results in the fields of computer vision and remote sensing. Unlike the above-mentioned methods, deep learning-based methods enhance the discrimination of backgrounds and anomalous targets by exploiting potential the high-level features in HSI. For HAD, unsupervised deep learning methods have drawn wide attention in anomaly detection. Autoencoder (AE) and generative adversarial network (GAN) are two common networks for HAD. Jiang et al. [32] proposed weakly supervised discriminative learning with spectrally constrained GANs for HAD. AE-based method detects anomaly by encoding and decoding a pixel, and the difference between the original input pixel and the decoded pixel is considered as the anomaly level of a pixel. Robust graph AE (RGAE) [33] was developed to improve the robustness of the network against anomalous targets and noise by embedding hypergraphs. Xiang et al. [34] proposed a HAD with guided AE (GAED) to reduce the feature representation of anomalous targets.

Currently, both representation-based methods and deep learning-based methods have achieved satisfactory performance. However, most of the current studies do not consider the highly mixed feature of pixels in HSIs. This makes the dictionary construction of representation-based methods have large errors and also prevents AE-based methods from adequately separating backgrounds and anomalies. Nonnegative matrix factorization (NMF) [35] is an important hyperspectral unmixing algorithm. It also was introduced in HAD to propose a similarity constrained convex NMF (SC-CNMF) method [36]. In HSI processing, endmembers and abundance are important concepts for interpreting spectral information and material composition in an image. Endmembers are pure spectral features in HSIs that represent different pure materials or substances. Abundance is the amount of presence or contribution of each endmember on each pixel, which is used to describe the distribution of different materials in an image. The endmembers in this method are acquired by convex NMF (CNMF) [37]. SC-CNMF method was first proposed for anomaly detection using endmembers. It uses the endmembers to construct the basis matrix that represents each pixel, and the reconstruction error is used to detect anomalies by means of a double window. However, the endmembers from SC-CNMF do not consider the representation properties of HSI. This means that the endmembers maybe not better reveal the intrinsic properties of HSI, which will influence the representation of anomalous targets to limit the detection performance. In other words, SC-CNMF just proves that NMF for anomaly detection is possible. But there is no better way to adapt to the properties of anomaly detection.

To better obtain the endmembers for HAD, we propose a novel method called learnable background endmember with subspace representation (LEBSR). It aims simultaneously to obtain pure endmembers and accurate abundances by unmixing HSIs. Unlike the simple unmixing work, we want to obtain endmembers that are more representative of background landcovers. Therefore, we use the priori properties to constrain the representation of background endmembers, which can better reveal the intrinsic background characteristics of HSI. Meanwhile, the p-norm constraint on the abundance is used to improve the representation accuracy of each pixel. Different from the other methods that start from the unmixing or endmember perspective, we make full use of the intrinsic representation of HSI to perform anomaly detection by constructing more accurate background endmembers.

The main highlights and contributions of this work can be briefly described as follows.

1) A LEBSR method is proposed for HAD, which can decompose HSI to obtain intrinsic background endmembers and accurate abundances. An anomaly detector is constructed based on the background endmembers and the abundances, which can better reconstruct the backgrounds while not for anomalies.
2) We consider the low-rank properties of backgrounds to achieve better representation of background

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endmembers in endmember subspace. We project the pre-obtained backgrounds into endmember subspace to facilitate the intrinsic representation of background endmembers.

3) To enhance the representation accuracy from background endmembers, we add the p-norm constraint on the abundance to better reveal the endmember component of each background pixel. This results in more accurate abundance representation.

The rest of this article is organized as follows. Section II describes the proposed LEBSR method in detail. In Section III, experiments are conducted and analyzed. Finally, Section IV concludes this article.

II. Proposed Method

A. Problem Formulation

According to the linear mixture model (LMM) [38], [39], the observed HSI $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^N \in \Re^D$ has N pixels with D spectral bands. For a pixel, it observes a spectrum consisting of a few pure spectral features of the landcovers called endmembers. A pixel in an HSI can be represented as

$$\mathbf{x}_i = \sum_{j=1}^{e} s_{ji}\mathbf{a}_j + \mathbf{n}_i \tag{1}$$

where e denotes the number of endmembers in the scene, $\{\mathbf{a}_j\}_{j=1}^e$ is the eth endmember, $\{s_{ji}\}_{j=1}^e$ corresponds to the abundance of the jth endmember in the ith pixel, $\mathbf{s}_i = [s_{1i}, s_{2i}, \ldots, s_{ei}]^T \in \Re^e$ is the abundance vector of $\mathbf{x}_i$, and $\mathbf{n}_i$ is the additive noise term.

Since the physical properties on the endmembers and the abundances in HSI have to satisfy the nonnegative condition. Therefore, (1) can be rewritten as follows:

$$\mathbf{x}_i = \sum_{j=1}^{e} s_{ji}\mathbf{a}_j + \mathbf{n}_i \quad \text{s.t. } \mathbf{a}_j \geq 0,\ \mathbf{s}_j \geq 0. \tag{2}$$

To obtain the endmembers and the abundances, a reconstructed model can be designed to decompose the HSI $\mathbf{X}$. According to model (1), we have

$$\min \frac{1}{2}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \sum_{j=1}^{e}\mathbf{s}_{ji}\mathbf{a}_j\right\|_2^2 \quad \text{s.t. } \mathbf{a}_j \geq 0, \quad \mathbf{s}_j \geq 0. \tag{3}$$

We can use the NMF method to solve for (3). Since the solution of NMF is affected by the initialization, to improve the robustness of the anomaly detector for HSIs, the abundance matrix also needs to satisfy a physical condition, i.e., the sum-to-one constraint. In addition, in HSI, a pixel does not generally contain all the endmembers. Generally, only marginal pixels consist of several endmembers, while other pixels just have one or two endmembers. The value of abundance for not containing endmember is zero. In the abundance $\mathbf{s}_i$, each column represents the abundance of a pixel and only a few entries are non-zero. Therefore, a sparsity constraint should be imposed on the abundance $\mathbf{s}_i$, which will improve the estimation accuracy of abundances.

In computer vision for sparsity constraint, p-norm minimization usually achieves better results than l_1-norm minimization, where p value can be set to 0–1 for the sparsity solution. In fact, both theoretical analysis and numerical experiments [40] demonstrate that p-norm ($0<p<1$) is close to l_1 minimization and more sparse. We have

$$\|\mathbf{s}_i\|_p^p = \sum_{j}^{e} |s_{ji}|^p. \tag{4}$$

Then, we can obtain the following objective function:

$$\min \frac{1}{2}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \sum_{j=1}^{e}\mathbf{s}_{ji}\mathbf{a}_j\right\|_2^2 + \alpha\sum_{i}^{N}\sum_{j}^{e}|s_{ji}|^p \quad \text{s.t. } \mathbf{a}_j \geq 0, \quad \mathbf{s}_j \geq 0, \ \sum_{j=1}^{e}\mathbf{s}_{ji} = 1 \tag{5}$$

where α is a tradeoff parameter of the sparsity constraint.

B. Learnable Background Endmember With Subspace Representation

In (5), the endmember and the abundance are solved simultaneously. To accurately reconstruct the backgrounds, the extracted endmembers should be close to the pure background endmembers.

To obtain good background endmembers, we introduce the priori background information to guide the representation of endmembers in the endmember subspace, and we use an approximation term for the endmembers and the priori backgrounds. Specifically, according to [24], we can obtain the priori backgrounds $\mathbf{D_0} = [\mathbf{d}_1, \mathbf{d}_2, \ldots, \mathbf{d}_{N_b}] \in \Re^{D\times N_b}$ with N_b atoms. The priori backgrounds should be well represented by the background endmembers in the endmember subspace. First, the priori backgrounds should be projected into the endmember subspace with a mapping matrix $\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_{N_b}]^T \in \Re^{N_b\times e}$. Then, the priori backgrounds can be represented with the coefficients $\mathbf{L} = [\mathbf{l}_1, \mathbf{l}_2, \ldots, \mathbf{l}_e]^T \in \Re^{e\times e}$ as follows:

$$\sum_{j=1}^{N_b}\left\|\mathbf{d}_j\mathbf{W} - \sum_{i=1}^{e}\mathbf{a}_i l_{ji}\right\|_2^2 \tag{6}$$

where $\mathbf{l}_i = [l_{1i}, l_{2i}, \ldots, l_{ei}]^T \in \Re^e$. Equation (7) can be rewritten as follows:

$$\min \frac{1}{2}\sum_{i=1}^{N}\left\|\mathbf{x}_i - \sum_{j=1}^{e}\mathbf{s}_{ji}\mathbf{a}_j\right\|_2^2 + \alpha\sum_{i}^{N}\sum_{j}^{e}|s_{ji}|^p + \lambda\sum_{j=1}^{N_b}\left\|\mathbf{d}_j\mathbf{W} - \sum_{i=1}^{e}\mathbf{a}_i l_{ji}\right\|_2^2 \quad \text{s.t. } \mathbf{a}_j \geq 0, \quad \mathbf{s}_j \geq 0, \ \sum_{j=1}^{e}\mathbf{s}_{ji} = 1 \tag{7}$$

where λ is a trade-off parameter.

The background endmembers $\mathbf{a}_j$ obtained from NMF can approximately represent the priori background information through $\mathbf{l}_i$, which allows the priori background information to

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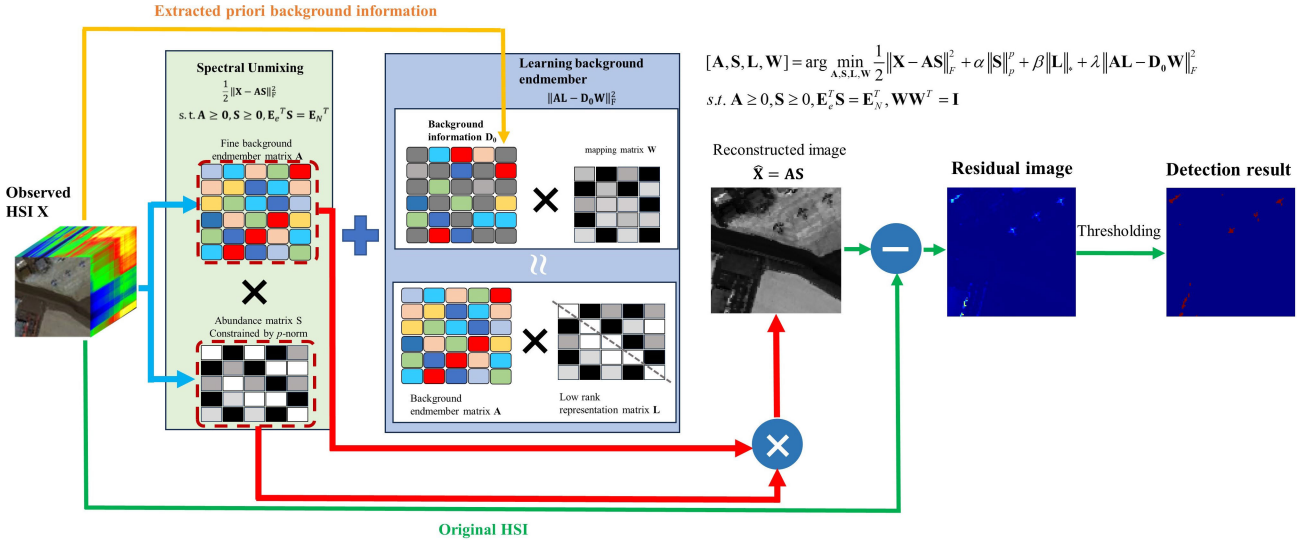


Fig. 1. Illustration of the proposed LEBSR method.

guide the learning of background endmembers. Specifically, $\mathbf{W}$ is a mapping matrix, the endmember $\mathbf{a}_j$ and the priori background information are in different spaces, $\mathbf{d}_j$ is mapped to the space of endmembers through $\mathbf{W}$, and the endmember $\mathbf{a}_j$ is approximately represented through $\mathbf{l}_i$ to let the two approximate. In other words only with the addition of the approximation term, the endmembers decomposed by LEBSR can be called background endmembers. In the endmember subspace, since the priori backgrounds and their projections contain a lot of redundant information, the mapped priori backgrounds are represented by the background endmembers in the term of low-rank.

Based on the above analysis, the matrix form of (7) can be obtained and its optimization problem is as follows:

$$\begin{aligned}[\mathbf{A},\mathbf{S},\mathbf{L},\mathbf{W}] = \arg\min_{\mathbf{A},\mathbf{S},\mathbf{L},\mathbf{W}} &\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2 + \alpha\|\mathbf{S}\|_p^p \\ &+ \beta\|\mathbf{L}\|_* + \lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2 \\ \text{s.t. } \mathbf{A}\geq 0,\quad &\mathbf{S}\geq 0,\ \mathbf{E}_e^T\mathbf{S}=\mathbf{E}_N^T,\ \mathbf{WW}^T=\mathbf{I}\end{aligned} \tag{8}$$

where $\mathbf{E}_e$ and $\mathbf{E}_N$ are the column vector with all elements equal to one. β is a trade-off parameter. $\|\cdot\|_*$ denotes the nuclear norm of a matrix. $\|\mathbf{L}\|_*$ is used to characterize the low-rank properties of representation coefficients. The orthogonality constraint $\mathbf{WW^T} = \mathbf{I}$ is added to obtain a closed solution for the mapping matrix $\mathbf{W}$.

C. LEBSR-Based Anomaly Detection

According to obtain the background endmembers and the abundances, we can reconstruct the HSI $\hat{\mathbf{X}}$ in the following:

$$\hat{\mathbf{X}} = \mathbf{AS}. \tag{9}$$

Since the background endmembers reveal the properties of backgrounds, they can be used to accurately represent the backgrounds, while they do not well represent the anomalies. Therefore, we can use the representation properties to discriminate the backgrounds and the anomalies. Specifically, we utilize the difference between the original image $\mathbf{X}$ and the reconstructed image $\hat{\mathbf{X}}$. Then we can obtain the residual image $\mathbf{r}$, where the backgrounds are suppressed and the anomalies are highlighted. The residual image can be obtained by

$$\mathbf{r} = \mathbf{X} - \hat{\mathbf{X}}. \tag{10}$$

According to the residual values, we can detect the anomalous pixels with a threshold value. In real scene, HSI generally contains noise to interfere the detection of anomalous pixels. Thus, to better highlight the detection of anomalous targets, we further optimize the detector, and the reconstructed residual $\gamma(x_i)$ of the ith pixel is defined as

$$\gamma_{(x_i)} = \left\|\mathbf{r}_{(:,i)} - \mathbf{u_s}\right\|_2^2 \tag{11}$$

where $\mathbf{r} = \mathbf{X}-\hat{\mathbf{X}}$ is the residual matrix, $\|[\cdot]_{:,i}\|_2$ is the ℓ_2-norm of the ith column of the residual matrix. $\mathbf{u_s}$ is the mean feature for all pixels in $\mathbf{r}$. If the residual value of $\mathbf{x}_i$ is greater than a threshold value, $\mathbf{x}_i$ is an anomalous pixel, otherwise, it is a background pixel. Algorithm 1 describes the AD of LEBSR in detail.

The whole process can be described in Fig. 1. An HSI $\mathbf{X}$ is input, and the original image is unmixed to obtain the

Algorithm 1 LEBSR-Based Anomaly Detection

Input: Background endmember matrix $\mathbf{A} \in \Re^{D\times e}$, abundance matrix $\mathbf{S} \in \Re^{e\times N}$, the ith detection pixel x_i, threshold value δ

Output: The anomaly detection result $\mathbf{Y}$

```
Initialize Y = 0, 0 ∈ ℜ^{1×N}
for i = 1 to N do
    Calculate γ_(x_i) using (11)
    if γ_(x_i) > δ then
        x_i is an anomalous pixel, Y(1, i) = 1
    end if
end for
```

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abundance matrix $\mathbf{S}$ and the background endmembers $\mathbf{A}$, where the abundance receives two physical constraints and a sparsity constraint with p-norm. Meanwhile, we use a priori background information $\mathbf{D_0}$ to guide the background endmembers, which is mapped into the endmember subspace based on an orthogonal basis $\mathbf{W}$. The background endmembers can be used to approximate the priori subspace information $\mathbf{D_0W}$ with a representation matrix $\mathbf{L}$. With the priori guidance, the background endmembers $\mathbf{A}$ can better represent the properties of backgrounds, and the abundances can be more accurate. Finally, the reconstructed image $\hat{\mathbf{X}}$ can be achieved according to the background endmembers $\mathbf{A}$ and the abundance matrix $\mathbf{S}$, and the residuals are generated with the original image $\mathbf{X}$ to obtain the final detection results. Compared with the original image, the residual image suppresses the backgrounds and highlights the anomalies.

D. Model Optimization

First, we can rewrite (8) as the following optimization problem:

$$\begin{aligned}\min_{\mathbf{A},\mathbf{S},\mathbf{L},\mathbf{W}} &\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2+R_+(\mathbf{A})+R_+(\mathbf{S})+\alpha\|\mathbf{S}\|_p^p\\&+\beta\|\mathbf{L}\|_*+\lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2+\frac{1}{2}\left\|\mathbf{E}_e^T\mathbf{S}-\mathbf{E}_N^T\right\|_F^2\\&+\frac{1}{2}\|\mathbf{WW}^T-\mathbf{I}\|_F^2 \end{aligned}\tag{12}$$

where $R_+(\cdot)$ is the equivalent form of the nonnegative constraint referring to [41]. To optimize the problem in (12), four auxiliary variables $\mathbf{P}$, $\mathbf{Z}$, $\mathbf{J}$, and $\mathbf{Q}$ are introduced to separate the variables, so it can be transformed as follows:

$$\begin{aligned}\min_{\mathbf{A},\mathbf{S},\mathbf{L},\mathbf{W}} &\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2+R_+(\mathbf{P})+R_+(\mathbf{Z})+\alpha\|\mathbf{J}\|_p^p\\&+\beta\|\mathbf{Q}\|_*+\lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2+\frac{1}{2}\left\|\mathbf{E}_e^T\mathbf{S}-\mathbf{E}_N^T\right\|_F^2\\&+\frac{1}{2}\|\mathbf{WW}^T-\mathbf{I}\|_F^2\\\text{s.t. }&\mathbf{P}=\mathbf{A},\quad \mathbf{Z}=\mathbf{S},\ \mathbf{J}=\mathbf{S},\ \mathbf{Q}=\mathbf{L}.\end{aligned}\tag{13}$$

Therefore, (13) can be solved by ADMM [42], and the augmented Lagrangian functions are expressed as

$$\begin{aligned}&\mathcal{L}_\mu(\mathbf{A},\mathbf{S},\mathbf{L},\mathbf{W},\mathbf{P},\mathbf{Z},\mathbf{J},\mathbf{Q},\mathbf{Y}_1,\mathbf{Y}_2,\mathbf{Y}_3,\mathbf{Y}_4,\mu)\\&=\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2+R_+(\mathbf{P})+R_+(\mathbf{Z})+\alpha\|\mathbf{J}\|_p^p\\&+\beta\|\mathbf{Q}\|_*+\lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2+\frac{1}{2}\left\|\mathbf{E}_e^T\mathbf{S}-\mathbf{E}_N^T\right\|_F^2\\&+\frac{1}{2}\|\mathbf{WW}^T-\mathbf{I}\|_F^2+\langle\mathbf{Y}_1,\mathbf{A}-\mathbf{P}\rangle+\frac{\mu}{2}\|\mathbf{A}-\mathbf{P}\|_F^2\\&+\langle\mathbf{Y}_2,\mathbf{S}-\mathbf{Z}\rangle+\frac{\mu}{2}\|\mathbf{S}-\mathbf{Z}\|_F^2+\langle\mathbf{Y}_3,\mathbf{S}-\mathbf{J}\rangle\\&+\frac{\mu}{2}\|\mathbf{S}-\mathbf{J}\|_F^2+\langle\mathbf{Y}_4,\mathbf{L}-\mathbf{Q}\rangle+\frac{\mu}{2}\|\mathbf{L}-\mathbf{Q}\|_F^2\end{aligned}\tag{14}$$

where the matrix inner product $\langle\mathbf{A},\mathbf{B}\rangle=\text{trace}(\mathbf{A}^T\mathbf{B})$, $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, and $\mathbf{Y}_4$ are Lagrange multipliers, and $\mu>0$ is a penalty parameter. The optimization problem can be decomposed into the following subproblems.

1) Updating $\mathbf{A}$ by fixing the other variables and solving the following problem:

$$\begin{aligned}\mathbf{A}=\underset{\mathbf{A}}{\text{argmin}}\ &\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2+\lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2\\&+\langle\mathbf{Y}_1,\mathbf{A}-\mathbf{P}\rangle+\frac{\mu}{2}\|\mathbf{A}-\mathbf{P}\|_F^2.\end{aligned}\tag{15}$$

2) Updating $\mathbf{S}$ by fixing the other variables and solving the following problem:

$$\begin{aligned}\mathbf{S}=\underset{\mathbf{S}}{\text{argmin}}\ &\frac{1}{2}\|\mathbf{X}-\mathbf{AS}\|_F^2+\frac{1}{2}\left\|\mathbf{E}_e^T\mathbf{S}-\mathbf{E}_N^T\right\|_F^2\\&+\langle\mathbf{Y}_2,\mathbf{S}-\mathbf{Z}\rangle+\frac{\mu}{2}\|\mathbf{S}-\mathbf{Z}\|_F^2\\&+\langle\mathbf{Y}_3,\mathbf{S}-\mathbf{J}\rangle+\frac{\mu}{2}\|\mathbf{S}-\mathbf{J}\|_F^2.\end{aligned}\tag{16}$$

3) Updating $\mathbf{P}$ by fixing the other variables and solving the following problem:

$$\mathbf{P}=\underset{\mathbf{P}}{\text{argmin}}\ R_+(\mathbf{P})+\langle\mathbf{Y}_1,\mathbf{A}-\mathbf{P}\rangle+\frac{\mu}{2}\|\mathbf{A}-\mathbf{P}\|_F^2.\tag{17}$$

4) Updating $\mathbf{Z}$ by fixing the other variables and solving the following problem:

$$\mathbf{Z}=\underset{\mathbf{Z}}{\text{argmin}}\ R_+(\mathbf{Z})+\langle\mathbf{Y}_2,\mathbf{S}-\mathbf{Z}\rangle+\frac{\mu}{2}\|\mathbf{S}-\mathbf{Z}\|_F^2.\tag{18}$$

5) Updating $\mathbf{J}$ by fixing the other variables and solving the following problem:

$$\mathbf{J}=\underset{\mathbf{J}}{\text{argmin}}\ \alpha\|\mathbf{J}\|_p^p+\langle\mathbf{Y}_3,\mathbf{S}-\mathbf{J}\rangle+\frac{\mu}{2}\|\mathbf{S}-\mathbf{J}\|_F^2.\tag{19}$$

6) Updating $\mathbf{Q}$ by fixing the other variables and solving the following problem:

$$\mathbf{Q}=\underset{\mathbf{Q}}{\text{argmin}}\ \beta\|\mathbf{Q}\|_*+\langle\mathbf{Y}_4,\mathbf{L}-\mathbf{Q}\rangle+\frac{\mu}{2}\|\mathbf{L}-\mathbf{Q}\|_F^2.\tag{20}$$

7) Updating $\mathbf{L}$ by fixing the other variables and solving the following problem:

$$\begin{aligned}\mathbf{L}=\underset{\mathbf{L}}{\text{argmin}}\ &\lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2+\langle\mathbf{Y}_4,\mathbf{L}-\mathbf{Q}\rangle\\&+\frac{\mu}{2}\|\mathbf{L}-\mathbf{Q}\|_F^2.\end{aligned}\tag{21}$$

8) Updating $\mathbf{W}$ by fixing the other variables and solving the following problem:

$$\mathbf{W}=\underset{\mathbf{W}}{\text{argmin}}\ \lambda\|\mathbf{AL}-\mathbf{D_0W}\|_F^2+\frac{1}{2}\|\mathbf{WW}^T-\mathbf{I}\|_F^2.\tag{22}$$

The solutions of the LEBSR model are outlined in Algorithm 2, where $\Theta_{\beta/\mu}$ and $\mathcal{S}_{\text{GST}}$ are the singular value threshold, and the generalized soft-thresholding function [40], [42], [43], respectively.

The main computational costs of algorithm 2 are in matrix inversion, matrix multiplication, singular value decomposition (SVD), and number of iterations. The computational cost of matrix addition is ignored to illustrate the computational complexity of this algorithm. In Algorithm 2, the main computational cost of solving $\mathbf{A}$ is about $\mathcal{O}(eDN_b+2e^3+e^2D+eDN+e^2D)$. The main computational cost of solving $\mathbf{S}$ requires about $\mathcal{O}(e^2D+e^2+e^3+eDN+eN)$. The computational cost of solving for $\mathbf{P}$ and $\mathbf{Z}$ can be neglected. We update

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Algorithm 2 Solving LEBSR Model

Input: HSI dataset $\mathbf{X} \in \Re^{D\times N}$, initialization background information $\mathbf{D_0} \in \Re^{D\times N_b}$;
$\mathbf{A} = \mathbf{S} = \mathbf{L} = \mathbf{0},\ \mathbf{P} = \mathbf{Z} = \mathbf{J} = \mathbf{Q} = \mathbf{0}$

Initialize: $\mathbf{Y}_1 = \mathbf{Y}_2 = \mathbf{Y}_3 = \mathbf{Y}_4 = \mathbf{0}, \mathbf{W} = \mathbf{E}$
$\mu_{max} = 10^{12}, \mu = 10^{-4}, \rho = 1.1, \varepsilon = 10^{-7}$

While not converged do

1: Fix others and update $\mathbf{A}$ via

$$\mathbf{A} = (2\lambda\mathbf{D_0}\mathbf{W}\mathbf{L}\mathbf{L}^T + \mathbf{X}\mathbf{S}^T + \mu\mathbf{P} - \mathbf{Y}_1)(\mathbf{S}\mathbf{S}^T + \mu\mathbf{I} + 2\lambda\mathbf{L}\mathbf{L}^T)^{-1} \tag{23}$$

2: Fix others and update $\mathbf{S}$ via

$$\mathbf{S} = (\mathbf{A}^T\mathbf{A} + \mathbf{E}_e\mathbf{E}_e^T + 2\mu\mathbf{I})^{-1}(\mathbf{A}^T\mathbf{X} + \mathbf{E}_e\mathbf{E}_N^T + \mu\mathbf{Z} - \mathbf{Y}_2 + \mu\mathbf{J} - \mathbf{Y}_3) \tag{24}$$

3: Fix others and update $\mathbf{P}$ via

$$\mathbf{P} = \max(\mathbf{A} + \frac{\mathbf{Y}_1}{\mu}, 0) \tag{25}$$

4: Fix others and update $\mathbf{Z}$ via

$$\mathbf{Z} = \max(\mathbf{S} + \frac{\mathbf{Y}_2}{\mu}, 0) \tag{26}$$

5: Fix others and update $\mathbf{J}$ via

$$\mathbf{J} = \mathcal{S}_{GST}(\mathbf{S} + \frac{\mathbf{Y}_3}{\mu}, \frac{\alpha}{\mu}) \tag{27}$$

6: Fix others and update $\mathbf{Q}$ via

$$\mathbf{Q} = \Theta_{\beta/\mu}\left(\mathbf{L} + \frac{\mathbf{Y}_4}{\mu}\right) \tag{28}$$

7: Fix others and update $\mathbf{L}$ via

$$\mathbf{L} = (2\lambda\mathbf{A}^T\mathbf{A} + \mu\mathbf{I})^{-1}(2\lambda\mathbf{A}^T\mathbf{D_0}\mathbf{W} + \mu\mathbf{J} - \mathbf{Y}_4) \tag{29}$$

8: Fix others and update $\mathbf{W}$ via

$$\mathbf{W} = (2\lambda\mathbf{D_0}^T\mathbf{D_0} + \mathbf{I})^{-1}(2\lambda\mathbf{D_0}^T\mathbf{A}\mathbf{L}) \tag{30}$$

9: Fix others and update $\mathbf{Y}_1$, $\mathbf{Y}_2$, $\mathbf{Y}_3$, $\mathbf{Y}_4$ by

$$\begin{aligned}\mathbf{Y}_1^* &= \mathbf{Y}_1 + \mu(\mathbf{A} - \mathbf{P})\\ \mathbf{Y}_2^* &= \mathbf{Y}_2 + \mu(\mathbf{S} - \mathbf{Z})\\ \mathbf{Y}_3^* &= \mathbf{Y}_3 + \mu(\mathbf{S} - \mathbf{J})\\ \mathbf{Y}_4^* &= \mathbf{Y}_4 + \mu(\mathbf{L} - \mathbf{Q})\end{aligned} \tag{31}$$

10: Update μ by

$$\mu^* = min(\rho\mu, \mu_{max}) \tag{32}$$

11: Check the convergence conditions

$$\begin{aligned}\|\mathbf{A} - \mathbf{P}\|_\infty &< \varepsilon\\ \|\mathbf{S} - \mathbf{Z}\|_\infty &< \varepsilon\\ \|\mathbf{S} - \mathbf{J}\|_\infty &< \varepsilon\\ \|\mathbf{L} - \mathbf{Q}\|_\infty &< \varepsilon\end{aligned} \tag{33}$$

End While

Output: A, S, L, W.

$\mathbf{Q}$ using SVD, and the complexity of solving $\mathbf{Q}$ is about $\mathcal{O}(e^3)$. Updating $\mathbf{L}$ and $\mathbf{W}$ needs about $\mathcal{O}(eDN_b + e^2N_b + e^3)$. Generally, $e \ll D \ll N$ and $e \ll N_b \ll N$. Therefore, the total computational complexity of Algorithm 2 is about $\mathcal{O}(\omega eDN)$, where ω is the number of iterations.

III. Experimental Result and Analysis

In the experiments, we validate the performance of the LEBSR model on 12 real datasets in this section. All the experiments are implemented using MATLAB 2020a on an Intel Core i5-10400F CPU machine with 64-bit and 16 GB of random access memory (RAM). The datasets are shown in Fig. 2 including their pseudo-color images and corresponding ground truth.

A. Dataset Description

1) HYDICE Urban Dataset [30]: This dataset was captured by the Hyperspectral Digital Image Acquisition Experiment (HYDICE) on an aircraft platform with a spectral resolution of 10 nm and a spatial resolution of about 1 m. The spatial size is 80 × 100 and has 175 spectral bands, with 162 bands remaining after removing low signal-to-noise ratio (SNR) and vapor absorption bands. There are 19 anomalous pixels in the HYDICE Urban dataset for detection.

2) Pavia Dataset [34]: This real data was acquired by an airborne sensor named the reflective optics system imaging spectrometer (ROSIS) with a spatial resolution of 1.3 m and a wavelength range of 430–860 nm. The image with 100 × 100 and 102 spectral bands has 71 anomalies.

3) MUUFL Dataset [34]: This HSI was collected at the University of Southern Mississippi Gulf Park Campus, Long Beach, MS, USA, by the hyperspectral compact airborne spectrographic imager (CASI-1500) [44]. The original HSI size is 325 × 220 × 72. The first four and the last four noise bands were removed in the experiments. A 200 × 220 scene was cropped from the original HSI. In this scene, four different colored cloth panels are considered anomalous targets.

4) Salinas Dataset [34]: The HSI was acquired by the AVIRIS at the Salinas Valley, CA, USA. A total of 512 × 217 pixels and 224 bands are included in this HSI, and its spatial resolution is 3.7 m. The low SNR and water absorption bands (108–112, 154–167, and 224) were removed. An HSI, containing 200 × 150 pixels, was cropped for experiments from the original HSI. The manufactured structures in this HSI are considered as anomalous targets.

5) SpecTIR Dataset: The experimental data were derived from the SpecTIR hyperspectral airborne Rochester experiment (SHARE) [45] with a 5 nm spectral resolution and 1 m ground spatial resolution. An image area of 180 ×180 pixels with 120 bands was used for evaluation. In this image scene, anomalies consist of different size man-made colorful square fabrics.

6) Gulfport Dataset [34]: This dataset was captured from the airborne visible/infrared imaging spectrometer (AVIRIS) sensor with a 3.4 m spatial resolution. The size of the HSI is 100 × 100. It has 191 spectral bands with the spectral wavelengths from 400 to 2500 nm after the bad bands are

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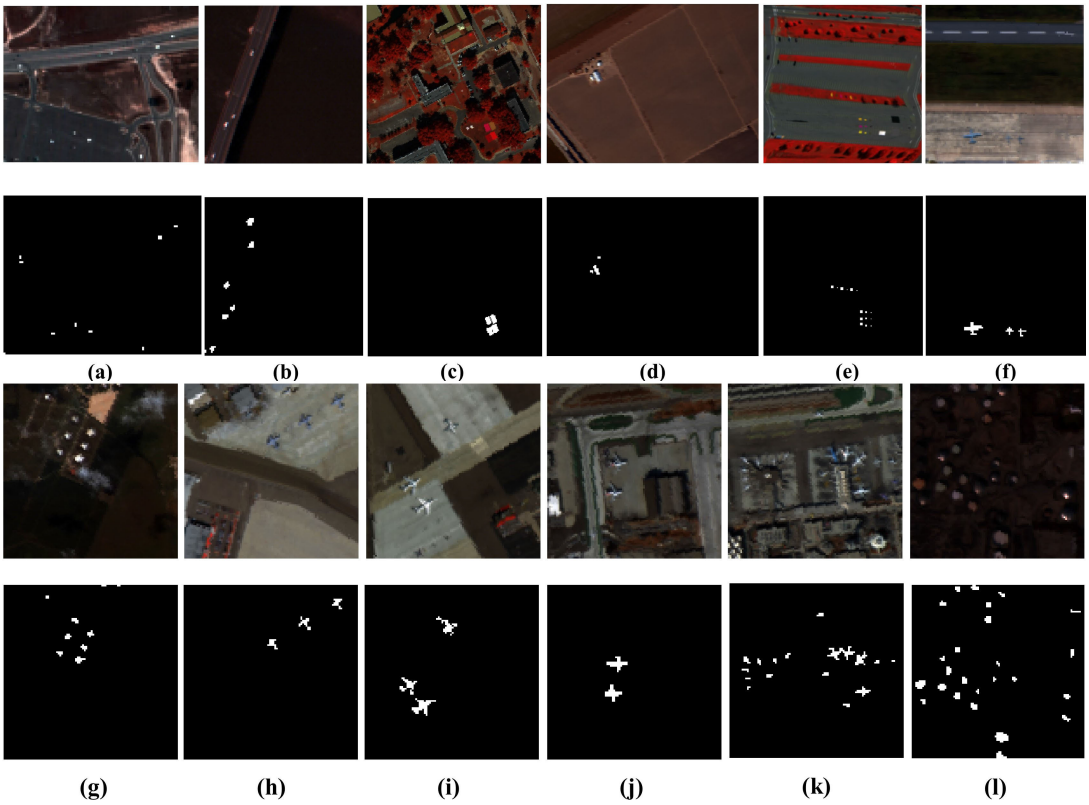


Fig. 2. Pseudo-color image and ground truth of experimental datasets. (a) HYDICE, (b) Pavia, (c) MUUFL, (d) Salians, (e) SpecTIR, (f) Gulfport, (g) Texas coast, (h) SD-1, (i) SD-2, (j) LA-1, (k) LA-2, and (l) LA-3.

removed because of the water absorption regions and low SNR. The types of anomalies in this HSI dataset are three airplanes at the bottom of the image, which occupies 60 pixels in total.

7) Texas Coast Dataset [46]: This urban dataset was captured by the AVIRIS airborne hyperspectral sensor over the Texas Coast, Texas, USA, in 2010. The spatial resolution and size of this scene are 17.2 m and 100 × 100 pixels. There are 67 anomalous pixels and 204 spectral bands in the wavelength range of 450–1350 nm.

8) San Diego (SD) Dataset [24]: These two HSIs were captured by the AVIRIS sensor in San Diego, California, USA with a spatial resolution of 3.5 m and a wavelength range of 370–2510 nm. The spatial size is 100 × 100 and has 189 spectral bands, which has removed the low SNR, absorbing regions, and bad bands. These two HSIs are called SD-1 and SD-2. The airports with 58 pixels in SD-1 and 134 pixels in SD-2 are considered as anomalies.

9) Los Angeles (LA) Dataset [46]: These three HSIs were acquired by the AVIRIS sensor covering the Los Angeles city area with a spatial resolution of 7.1 m. The spatial size is 100 × 100 and has 205 spectral bands, which has removed the low SNR, absorbing regions, and bad bands. These three HSIs are called LA-1, LA-2, and LA-3. The airports with 87 pixels in LA-1 and 170 pixels in LA-2 are considered as anomalies and the buildings with 272 pixels in LA-3 as anomalous targets.

B. Experimental Setup

In this article, a total of ten representative HAD methods are used for comparison, including the statistical-based methods such as global RX [12], the CR-based methods such as CRD [30] and RCRD [31], the NMF-based method such as CNMF [36], the SR- and LR-based methods such as LSMAD [23], LRASR [24], SaFra [29], and AHMID [26], and the AE-based methods such as RGAE [33] and GAED [34]. We did experiments on twelve datasets and showed the results in Table I.

In our proposed method, the p-norm is used to constrain the abundance matrix, and we analyze the influence of p with the values from zero to one. In our experiments, we take $p = 0.5$. For the number of background endmembers e, it represents the number of background endmember to be learned and we analyze its influence on detection results. In addition, the model needs to construct priori background information to guide the learning of background endmember, and we adopt the strategy in [24].

To measure the detection performance of different AD methods, the receiver operating characteristic (ROC) is widely used for intuitive observation. We use the ROC curves of (P_f, P_d) and (τ, P_f) to represent the overall detection accuracy and the background suppression ability, respectively. P_f and P_d are related to τ, and they can be obtained by

$$P_d^{(\tau)} = \frac{\mathrm{TP}^{(\tau)}}{N_{\mathrm{ano}}}, \quad P_f^{(\tau)} = \frac{\mathrm{FP}^{(\tau)}}{N_{\mathrm{bkg}}} \tag{34}$$

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TABLE I
AUC VALUES OF ALL COMPARED METHODS FOR THE DATASETS

$AUC_{(P_f,\ P_d)}$	RX	CRD	RCRD	LSMAD	LRASR	SaFra	AHMID	RGAE	GAED	CNMF	LEBSR
HYDICE	0.9824	0.9892	0.9989	0.9925	0.9846	0.9903	0.9985	0.9201	0.9985	0.9809	**0.9996**
Pavia	0.9934	0.9381	0.99251	0.9914	0.9263	0.9934	0.9949	0.9938	0.9948	0.9639	**0.9963**
MUUFL	0.9970	0.8562	0.9923	0.9975	0.9892	0.9184	0.9928	0.9843	0.9749	0.9859	**0.9981**
Salians	0.9994	0.9730	0.9994	0.9998	0.9998	0.9974	0.9998	0.9912	0.9999	0.9821	**0.9999**
SpecTIR	0.9914	0.9471	0.9116	0.9965	0.9207	0.9101	0.9680	0.9332	0.9694	0.9417	**0.9990**
GulfPort	0.9526	0.8378	0.9885	0.9869	0.9681	0.9457	0.9917	0.8932	0.9686	0.7828	**0.9923**
Texas Coast	0.9907	0.9655	0.9861	0.9840	0.9871	0.9785	0.9784	0.9468	0.9447	0.7633	**0.9935**
SD-1	0.8885	0.9432	0.8626	0.9763	0.9647	0.9901	0.9923	0.9824	0.9890	0.9616	**0.9944**
SD-2	0.9519	0.9613	0.9630	0.9777	0.9630	0.9817	0.9891	**0.9926**	0.9907	0.8468	0.9920
LA-1	0.8404	0.8840	0.9781	0.9176	0.7051	0.9477	0.9485	0.9519	0.9691	0.7906	**0.9801**
LA-2	0.9288	0.9098	0.9011	0.9379	0.9001	0.9348	0.9685	0.8971	0.8661	0.8786	**0.9742**
LA-3	0.9887	0.8973	0.9862	0.9806	0.9801	0.9873	**0.9964**	0.9943	0.9925	0.9631	0.9952
Average	0.9588	0.9252	0.9634	0.9782	0.9407	0.9646	0.9849	0.9567	0.9715	0.9034	**0.9929**
$AUC_{(\tau,\ P_f)}$	RX	CRD	RCRD	LSMAD	LRASR	SaFra	AHMID	RGAE	GAED	CNMF	LEBSR
HYDICE	0.0345	0.0339	0.1771	0.0170	0.0808	0.0604	0.0105	0.0770	0.0080	0.0542	**0.0040**
Pavia	0.0258	0.1088	0.0028	0.0140	0.0405	0.0294	0.0056	0.0050	0.0056	0.1000	**0.0012**
MUUFL	0.0237	0.0306	0.0840	0.0189	0.0574	0.1195	0.0101	0.0347	0.0232	0.0515	**0.0070**
Salians	0.0175	0.0790	0.0292	0.0207	0.0674	0.0150	0.0093	0.0138	**0.0031**	0.0176	0.0071
SpecTIR	0.0249	0.0462	0.1005	0.0123	0.1863	0.1736	0.0329	0.0222	0.0080	0.1392	**0.0054**
GulfPort	0.0248	0.0218	0.0149	0.0536	0.1086	0.0881	0.0350	0.1499	0.0345	0.1397	**0.0110**
Texas Coast	0.0556	0.0127	0.0210	0.0560	0.0827	0.0927	**0.0024**	0.0478	0.0635	0.1560	0.0102
SD-1	0.0381	0.0465	0.1903	0.0232	0.0546	0.0097	0.0080	0.0172	0.0100	0.0320	**0.0059**
SD-2	0.0707	0.0469	0.0146	0.0271	0.0872	0.0599	0.0060	0.0074	0.0086	0.0407	**0.0056**
LA-1	0.0516	0.0504	**0.0073**	0.0310	0.0703	0.0271	0.0161	0.0159	0.0182	0.0590	0.0078
LA-2	0.0145	0.0329	0.0269	0.0036	0.0171	0.0340	0.0141	0.0048	0.0015	0.0205	**0.0008**
LA-3	0.0114	0.0015	0.0014	0.0017	0.0126	0.0021	0.0004	0.0004	0.0005	0.0233	**0.0003**
Average	0.0328	0.0426	0.0558	0.0233	0.0727	0.0593	0.0125	0.0330	0.0154	0.0695	**0.0055**

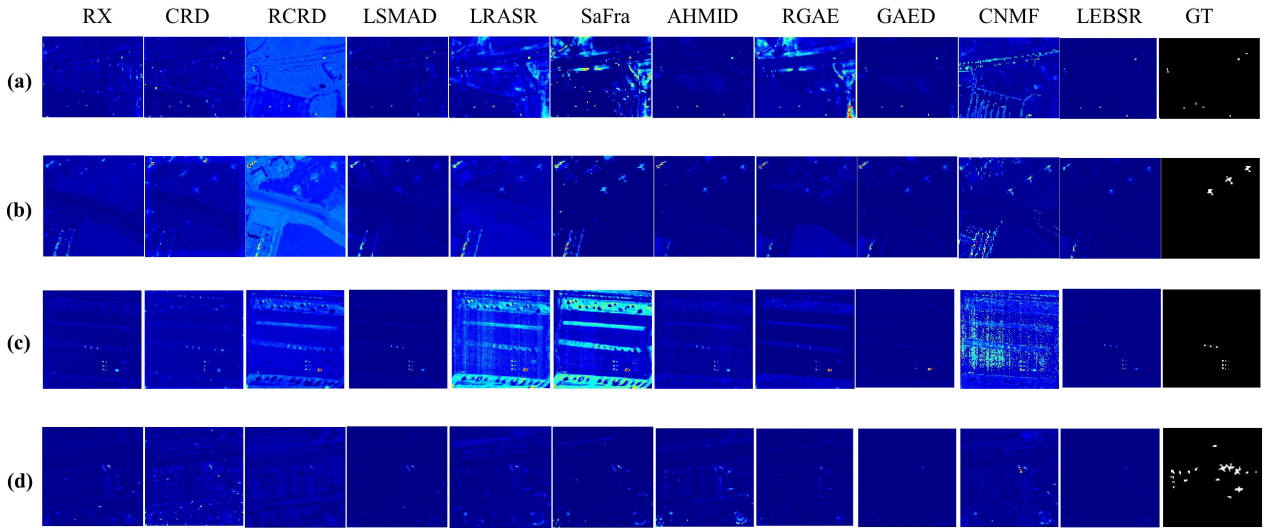


Fig. 3. Detection maps of different methods. (a) HYDICE, (b) SD-1, (c) SepcTIR, and (d) LA-2.

where $TP^{(\tau)}$ and $FP^{(\tau)}$ denote the number of anomalies and background pixels whose detection values are greater than a given threshold τ, N_{ano}, and N_{bkg} are the total number of anomalies and background pixels.

C. Experimental Results and Analysis

Table I shows the AUC values from the ROC curves. $\text{AUC}_{(P_f,P_d)}$ reflects the detection accuracy, while $\text{AUC}_{(\tau,P_f)}$ responds to the background suppression accuracy. From Table I, our proposed LEBSR method can achieve the highest score in bold on ten datasets for $\text{AUC}_{(P_f,P_d)}$ and the second position in underline on the other two datasets. For the $\text{AUC}_{(\tau,P_f)}$ of the proposed method, the smallest values are obtained on nine datasets and the other three datasets are the second smallest position. In addition, LEBSR has the best average score on all 12 datasets for $\text{AUC}_{(P_f,P_d)}$ and $\text{AUC}_{(\tau,P_f)}$, which is attributed to the fact that the proposed method can represent the background well via the learnable background endmember. Thus, it performs better detection rate and background suppression.

To qualitatively reflect the detection results, we selected four datasets including HYDICE, SD-1, SpecTIR, LA-2. Fig. 3 depicts the detection maps of different AD algorithms on the four HSI datasets. From Fig. 3, the RX method misses a lot of anomalous targets and misdetects a large number of backgrounds on the four datasets, which is very difficult to

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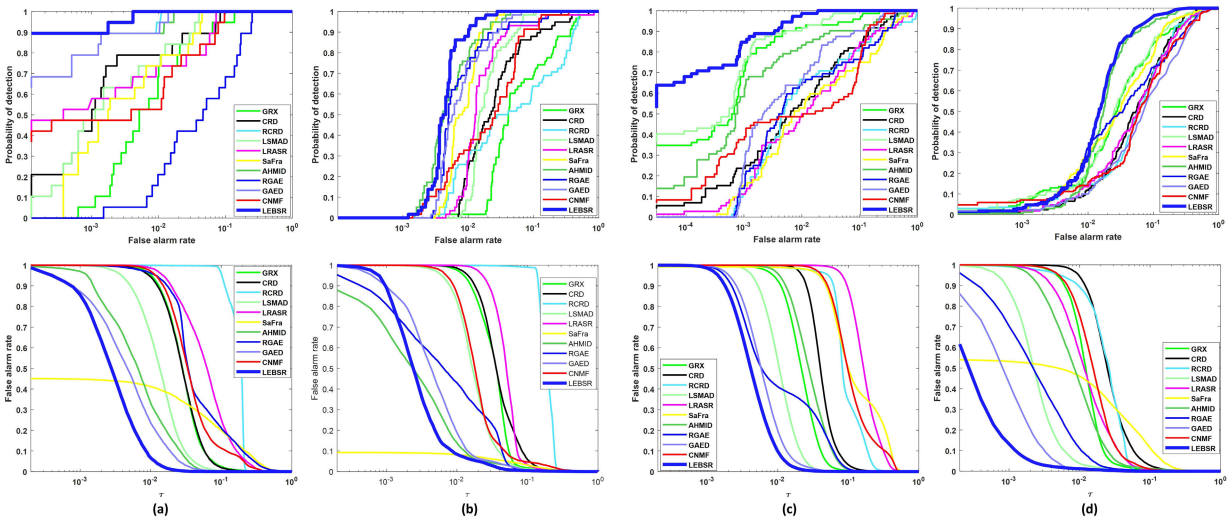


Fig. 4. ROC of (P_f, P_d) and (τ, P_f) for different methods. (a) HYDICE, (b) SD-1, (c) SepcTIR, and (d) LA-2.

separate the anomalies and backgrounds. Since RX is based on a simple statistical rule that is difficult to match the statistical distribution for complex HSI. The CRD and RCRD methods clearly display the backgrounds, resulting in the anomalies being difficultly discriminated from backgrounds. The LRASR and SaFra have a lot of false anomaly detection results. The LSMAD and AHMID methods can detect anomalous targets on all datasets, but the backgrounds are also relatively obvious. For the comparison of RGAE and GAED, they both can detect the anomalies while GAED achieves better background suppression. For the CNMF method, since the endmembers do not consider the background properties, its background suppression is poor. Compared to the other methods, the proposed LEBSR method successfully detects the anomalous targets, separates the anomalous targets from the backgrounds, and achieves satisfactory detection accuracy and background suppression.

To further analyze the detection performance of different methods, the two ROC curves are shown in Fig. 4. For better visualization, we show them on a logarithmic scale. The ROC curves of (P_f, P_d) and (τ, P_f) are near the upper left and lower the left corners, indicating that the detector is able to obtain high probability of detecting low false alarms, respectively. From the ROC curves of (P_f, P_d), LEBSR has greater detection probability than the other methods on the four datasets under different false alarm rates. For the LA-2 dataset, CNMF has the largest detection probability when the false alarm rate is less than 0.001, while LEBSR has the largest detection probability when the false alarm rate is greater than 0.001. Overall, the proposed LEBSR stays above the curves of other methods for the ROC curves of (P_f, P_d) and below the curves of other methods for the ROC curves of (τ, P_f), which means that our proposed method can maintain a high detection rate with a low false alarm rate on all the four datasets to show the consistent visual detection results as shown in Fig. 3.

D. Discussion and Analysis

1) Separability Analysis: To analyze the separability of background and anomalies, Fig. 5 shows the separation box-plots between anomalies and backgrounds according to the detection values, which reveals the separability of the anomalous targets from the background pixels to measure the detection performance. In the figure, the red box-plots mean the anomalous targets and the blue box-plots indicate the backgrounds. In general, these methods have different separation effects on different datasets. If the margin between the backgrounds and the anomalies is large, the anomalous pixels can be easily detected from the backgrounds, indicating that the detector has good performance. A large overlap between anomalies and backgrounds, such as LRASR and CNMF on the SpecTIR dataset, which indicates that the anomaly separation ability is limited. For the backgrounds, a small box-pot explains that the backgrounds are relatively concentrated, which indicates that the background suppression is very strong. SaFra shows strong separation on HYDICE and SD-1, but has significant limitations in separating background and anomalies on SpecTIR and LA-2. For LEBSR, there is almost no overlap between the backgrounds and the anomalies on the four datasets and the backgrounds are very compactness. The results show that the proposed method achieves strong background suppression and anomaly separation ability.

2) Ablation Studies: The proposed LEBSR method mainly includes two key constrains, i.e., p-norm (PN) for the abundance matrix, and subspace LRR for endmember backgrounds (LEB). We conduct the ablation analysis on the four datasets to validate the effectiveness of each part. Table II shows the results of the ablation experiments. The ablation methods achieve weaker results than the completed LEBSR in terms of $\mathrm{AUC}_{(P_f,\ P_d)}$ and $\mathrm{AUC}_{(\tau,\ P_f)}$. The removal of the PN module has a significant negative impact on the detection, especially on the $\mathrm{AUC}_{(\tau,\ P_f)}$, suggesting the PN model can extract more accurate abundance with p norm constraints. Similarly, the removal of the LEB module negatively affects the detection effectiveness and background suppression ability of the model, as the LEB module can further improve the representation of the background endmembers. These results show that the proposed model is more effective for both the anomaly detection and the background suppression.

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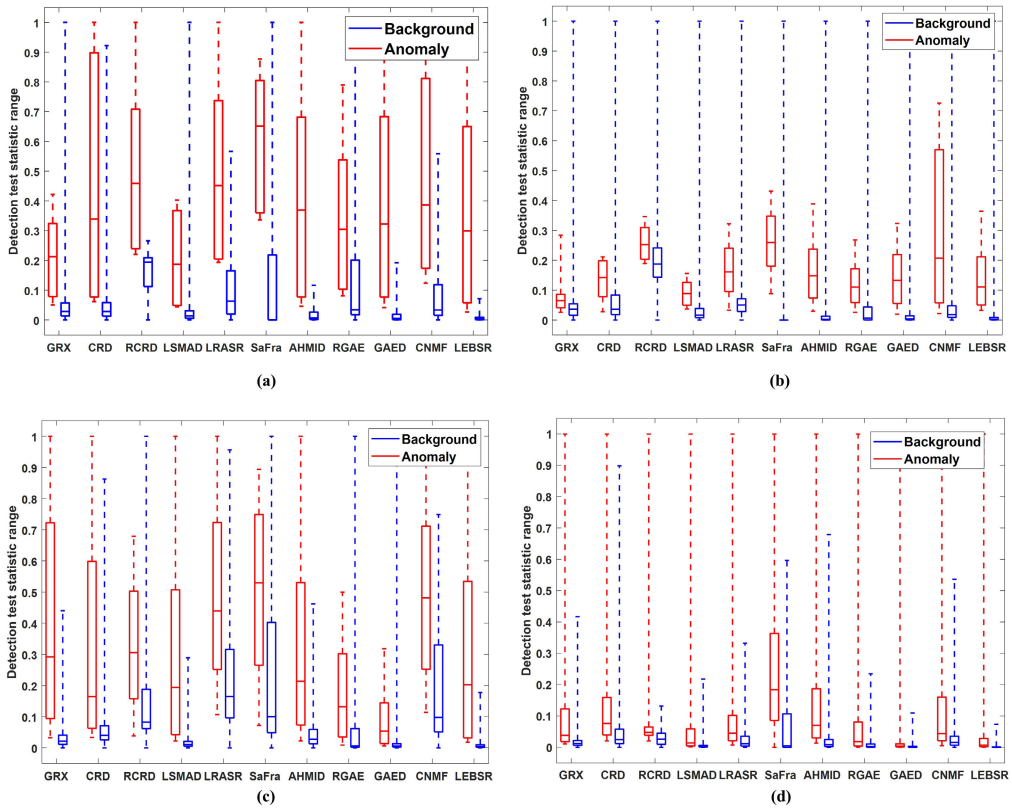


Fig. 5. Separation box-plots of different methods on the different datasets. (a) HYDICE, (b) SD-1, (c) SpecTIR, and (d) LA-2.

TABLE II
ABLATION EXPERIMENTS ON THE FOUR DATASET

LEBSR	$AUC_{(P_f,\ P_d)}$				$AUC_{(\tau,\ P_f)}$			
	HYDICE	SD-1	SpecTIR	LA-2	HYDICE	SD-1	SpecTIR	LA-2
w/o PN	0.9743	0.9338	0.8864	0.9427	0.0622	0.0273	0.0995	0.0563
w/o LEB	0.9918	0.9625	0.9941	0.9360	0.0081	0.0122	0.0107	0.0016
w/o Eq. (11)	0.9995	0.9842	0.9988	0.9741	0.0468	0.0569	0.0477	0.0161
w/o W	0.9714	0.9479	0.9898	0.9077	0.0795	0.0061	0.0057	0.0021
LEBSR	**0.9996**	**0.9944**	**0.9990**	**0.9742**	**0.0040**	**0.0059**	**0.0054**	**0.0008**

We further verify the necessity of (11) and the mapping matrix $\mathbf{W}$ in our LEBSR. From Table II, without (11), although the overall detection accuracy is not affected too much, the $\mathrm{AUC}_{(\tau,\ P_f)}$ value drops dramatically, which explains (11) is effective to suppress the background. Without the mapping matrix $\mathbf{W}$, the AUC values both decrease. This indicates that there is a large impact on the learning of background endmembers without mapping the priori backgrounds into the endmember space.

3) Analysis of Value p and Number of Endmembers e: As we mentioned in the model, in addition to the value p has an impact on the the estimation of abundances, and the number of endmembers e affects the description of backgrounds. We experimentally analyzed their impact on the detection performance. We set the value of p as 0.1, 0.3, 0.5, 0.7, and 0.9 since the range of p is 0–1 and e as 2, 4, 6, 8, and 10. Since the SpecTIR image size is larger and the backgrounds are more complex, the number of endmembers needs to be larger, and we set the e as 5, 10, 15, 20, and 25 on the SpecTIR dataset. The experimental results are shown in Fig. 6. The proposed LEBSR has the potential to obtain different AUC values for different values of p and e. For the value p, the robustness of LEBSR is affected when p is close to 0 and 1, and stronger detection performance and higher AUC

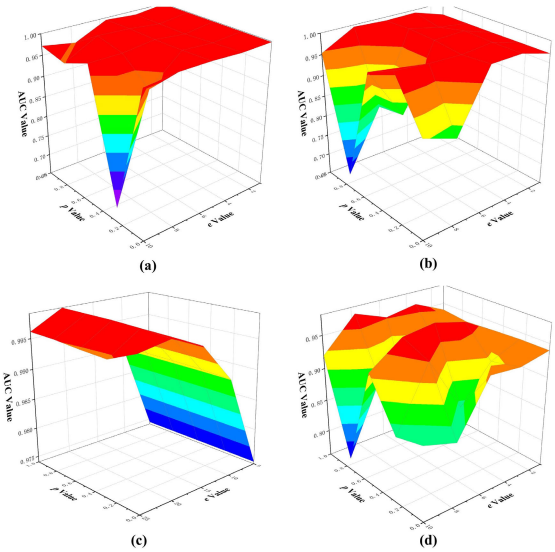


Fig. 6. AUC value performance with different values e and values p on the different datasets. (a) HYDICE, (b) SD-1, (c) SpecTIR, and (d) LA-2.

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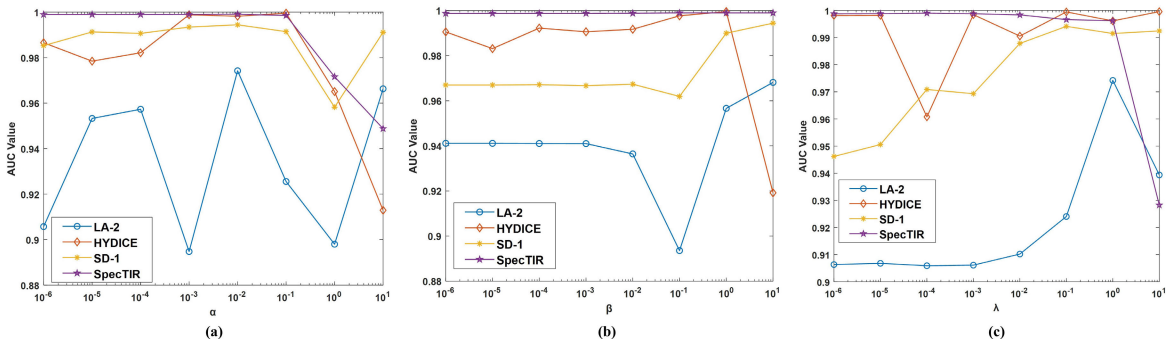


Fig. 7. AUC values of LEBSR with different parameters on the four datasets. (a) α, (b) β, and (c) λ.

TABLE III
RUNNING TIME OF DIFFERENT METHODS (S)

Dataset	RX	CRD	RCRD	LSMAD	LRASR	SaFra	AHMID	RGAE	GAED	CNMF	LEBSR
HYDICE	**0.05**	0.84	54.41	7.69	5.74	48.05	41.51	51.90	35.07	13.61	1.10
SD-1	**0.07**	15.89	42.16	12.85	6.88	68.53	45.23	125.77	53.55	20.57	1.78
SpecTIR	**0.12**	10.58	81.79	20.74	16.07	127.80	101.94	253.70	143.14	38.01	15.13
LA-2	**0.07**	1.30	116.44	13.09	14.78	48.68	152.70	129.26	53.94	22.03	2.06
Average	**0.08**	7.15	73.70	13.59	10.87	73.26	85.35	140.16	71.43	23.56	5.02

values can be obtained when p is 0.5. For the number of background endmembers e, taking more endmembers has a positive effect on the detection results. Since the background landcovers in HSI is finite, choosing too many endmembers will make the decomposed background endmembers contain anomalies, which will lead to a decrease in the detection accuracy. Therefore, a reasonable number of endmembers is very key to LEBSR. The background components of different HSI datasets are different, so we recommend that for different HSI size, we choose the corresponding number of endmemebrs e. The larger the HSI is, the more endmembers we choose. p is recommended to choose 0.5.

4) Parameter Analysis: LEBSR includes the three tradeoff parameters α, β, and λ. We evaluate the effectiveness of parameters on the four datasets in Fig. 7. We set the values of the parameters as 0.00001, 0.0001, 0.001, 0.01, 0.1, 1, and 10. From the AUC values under different values of α, β, and λ, the proposed LEBSR is more likely to obtain higher AUC values when the value of α is small. For the values of β and λ, more robust detection performance and higher AUC values are obtained when close to 1. As found in the experiments, we suggest setting α at [0.01, 0.1] and β and λ at [0.5, 1.5] for the LEBSR detector on the dataset presented in the article to obtain a robust solution with good detection performance. For new datasets, the characteristics of the data are specifically analyzed to adjust the hyperparameters.

5) Running Time Analysis: We also compare the running time of all the above methods on different datasets, and the detailed results are shown in Table III. RX is a simple model and performs more efficiently than other methods. The CRD method uses a double window to estimate the local background of one pixel under test and has higher computational complexity than RX. The RCRD method spends more time on dictionary construction. RGAE and GAED belong to multilayer learning and also take more time. Although our method also requires iterative optimization, its solution is in the subspace, which reduces a lot of time and is the second least time-consuming compared to the other methods.

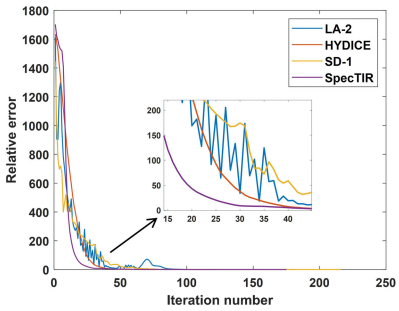


Fig. 8. Relative errors versus iterations on four datasets.

6) Convergence Analysis: The numerical convergence is investigated on the four datasets. Fig. 8 shows the relative error given in (33) versus the number of iterations. As the number of iterations increases, there are some fluctuations at first. In the subsequent iterations, the relative error decreases monotonically and finally approaches to zero.

IV. CONCLUSION

In this article, we propose an AD method for HSI, called LEBSR. Unlike previous HAD methods, the LEBSR method represents the backgrounds for anomaly detection by learning accurate abundances and background endmembers. The background endmembers and accurate abundances are directly applied to anomaly detection from the perspective of unmixing guided by the subspace representation of the priori background information. The constraint of p-norm and the learnable background endmembers strategy are used to obtain more accurate abundance as well as more representative background endmembers. Extensive experiments validate the advantages of the proposed method over many state-of-the-art methods on some benchmark datasets.

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